汪汗青 (Hanqing Wang)

Email: hanqingwang.c@gmail.com | Shanghai, Xuhui District, China | https://hanqingwangai.github.io

EDUCATION

B.S. in Computer Science — Sep 2013 - Jun 2017
Beijing Institute of Technology (BIT), Beijing, China

M.S. in Computer Science — Sep 2017 - Jun 2018
Beijing Institute of Technology (BIT), Beijing, China , ***Advisor***: *Prof. Wei Liang*

Ph.D. in Computer Science — Sep 2018 - Mar 2024
Beijing Institute of Technology (BIT), Beijing, China , ***Advisor***: *Prof. Wei Liang*

Visiting Ph.D. Student — Sep 2021 - Jun 2023
ETH Zurich, Zurich, Switzerland , ***Advisor***: *Prof. Luc Van Gool*

PROFESSIONAL EXPERIENCE

Research Intern — Mar 2017 - Nov 2017
Microsoft Research Asia (MSRA), Beijing, China, ***Advisor***: *Dr. Jiaolong Yang*
Research Experience: Single-View 3D Reconstruction

Research Intern — Jul 2019 - Apr 2020
Inception Institute of Artificial Intelligence (IIAI), Abu dhabi, UAE, ***Advisor***: *Prof. Wenguan Wang*
Research Experience: Embodied Vision-and-Language Navigation

Research Intern — Jul 2023 - Mar 2024
Shanghai Artificial Intelligence Laboratory, Shanghai, China, ***Advisor***: *Dr. Jiangmiao Pang*
Research Experience: Large-Scale Embodied Research Benchmark

Researcher & Postdoctoral Fellow — Apr 2024 - Nov 2025
Shanghai Artificial Intelligence Laboratory, Shanghai, China, ***Advisor***: *Prof. Dahua Lin*
Leader of the Digital Modeling and Simulation Platform Team, Embodied AI Center.
Research Experience: Simulation for Embodied AI, Vision-and-Language Navigation, Generalist Robot Policies

Research Scientist — Dec 2025 - Now
Shanghai Artificial Intelligence Laboratory, Shanghai, China,
Leader of the Digital Modeling and Simulation Platform Team, Embodied AI Center.
Research Experience: Simulation for Embodied AI, Vision-and-Language Navigation, Generalist Robot Policies

RESEARCH INTERESTS

- **Embodied Navigation & Perception** —Vision-and-Language Navigation, Active Perception, Exploration
- **Generalist Robot Policies** —Vision-Language-Action Models, Robotic Manipulation
- **Embodied Simulation & Benchmarks** —Large-scale Simulators, 3D Scene Understanding

SELECTED PROJECTS

- **InternUtopia** —Project Lead & Core Contributor. Large-scale simulation platform for embodied AI research. [github]
- **InternNav** —Project Lead & Core Contributor. Open platform for building generalized navigation foundation models. [github]

PROGRAMMING SKILLS

Programming Languages C/C++, Python, OpenUSD, LaTeX

Frameworks PyTorch, Hugging Face, IsaacSim

SELECTED PUBLICATIONS

† denotes corresponding author.

2026

Yunsong Zhou, Hangxu Liu, Xuekun Jiang, Xing Shen, Yuanzhen Zhou, Hui Wang, Baole Fang, Yang Tian, Mulin Yu, Qiaojun Yu, Li Ma, Hengjie Li, **Hanqing Wang**, Jia Zeng, Jiangmiao Pang, SIM1: Physics-Aligned Simulator as Zero-Shot Data Scaler in Deformable Worlds, *arXiv preprint arXiv:2604.08544*, *2026*

Ningyu Yan, Shuai Wang, Xing Shen, Hui Wang, **Hanqing Wang**†, Yang Xiang, Jiangmiao Pang, Tac2Real: Reliable and GPU Visuotactile Simulation for Online Reinforcement Learning and Zero-Shot Real-World Deployment, *arXiv preprint arXiv:2603.28475*, *2026*

Junhao Cai, Zetao Cai, Jiafei Cao, Yilun Chen, **Hanqing Wang**, et al., InternVLA-A1: Unifying Understanding, Generation and Action for Robotic Manipulation, *arXiv preprint arXiv:2601.02456*, *2026*

2025

Wensi Huang, Shaohao Zhu, Meng Wei, Jinming Xu, Xihui Liu, **Hanqing Wang**†, Tai Wang, Feng Zhao, Jiangmiao Pang, VL-LN Bench: Towards Long-horizon Goal-oriented Navigation with Active Dialogs, *arXiv preprint arXiv:2512.22342*, *2025*

Xinyi Chen, Yilun Chen, Yanwei Fu, Ning Gao, **Hanqing Wang**, et al., InternVLA-M1: A Spatially Guided Vision-Language-Action Framework for Generalist Robot Policy, *arXiv preprint arXiv:2510.13778*, *2025*

Weipeng Zhong, Peizhou Cao, Yichen Jin, Li Luo, Wenzhe Cai, Jingli Lin, **Hanqing Wang**, Zhaoyang Lyu, Tai Wang, Bo Dai, Xudong Xu, Jiangmiao Pang, InternScenes: A Large-Scale Simulatable Indoor Scene Dataset with Realistic Layouts, *arXiv preprint arXiv:2509.10813*, *2025*

Meng Wei, Chenyang Wan, Xiqian Yu, Tai Wang, Yuqiang Yang, Xiaohan Mao, Chenming Zhu, Wenzhe Cai, **Hanqing Wang**, Yilun Chen, Xihui Liu, Jiangmiao Pang, StreamVLN: Streaming Vision-and-Language Navigation via SlowFast Context Modeling, *arXiv preprint arXiv:2507.05240*, *2025*

Liuyi Wang, Xinyuan Xia, Hui Zhao, **Hanqing Wang**†, Tai Wang, Yilun Chen, Chengju Liu, Qijun Chen, Jiangmiao Pang, Rethinking the Embodied Gap in Vision-and-Language Navigation: A Holistic Study of Physical and Visual Disparities, *International Conference on Computer Vision (ICCV)*, *2025*

Boyu Mi, **Hanqing Wang**†, Tai Wang, Yilun Chen, Jiangmiao Pang, Language-to-Space Programming for Training-Free 3D Visual Grounding, *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, *2025*

Ning Gao, Yilun Chen, Shuai Yang, Xinyi Chen, Yang Tian, Hao Li, Haifeng Huang, **Hanqing Wang**, Tai Wang, Jiangmiao Pang, GenManip: LLM-driven Simulation for Generalizable Instruction-Following Manipulation, *Conference on Computer Vision and Pattern Recognition (CVPR)*, *2025*

2024

Hanqing Wang, Jiahe Chen, Wensi Huang, Qingwei Ben, Tai Wang, et al., GRUtopia: Dream General Robots in a City at Scale, *arXiv preprint arXiv:2407.10943*, *2024*

Dong An, **Hanqing Wang**, Wenguan Wang, Zun Wang, Yan Huang, Keji He, Liang Wang, ETPNav: Evolving Topological Planning for Vision-Language Navigation in Continuous Environments, *IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)*, *2024*

Zan Wang, **Hanqing Wang**, Wei Liang, Mastering Scene Rearrangement with Expert-Assisted Curriculum Learning and Adaptive Trade-Off Tree-Search, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, *2024*

2023

Hanqing Wang, Wei Liang, Luc Van Gool, Wenguan Wang, DREAMWALKER: Mental Planning for Continuous Vision-Language Navigation, *International Conference on Computer Vision (ICCV)*, *2023*

Chuanqi Zang, **Hanqing Wang**, Mingtao Pei, Wei Liang, Discovering the Real Association: Multimodal Causal Reasoning in Video Question Answering, *Conference on Computer Vision and Pattern Recognition (CVPR)*, *2023*

Hanqing Wang, Wenguan Wang, Wei Liang, Steven C. H. Hoi, Jianbing Shen, Luc Van Gool, Active Perception for Visual-Language Navigation, *International Journal of Computer Vision (IJCV)*, *2023*

Before 2023

Hanqing Wang, Wei Liang, Luc Van Gool, Wenguan Wang, Towards Versatile Embodied Navigation, *Conference on Neural Information Processing Systems (NeurIPS)*, *2022*

Hanqing Wang, Wei Liang, Jianbing Shen, Luc Van Gool, Wenguan Wang, Counterfactual Cycle-Consistent Learning for Instruction Following and Generation in Vision-Language Navigation, *Conference on Computer Vision and Pattern Recognition (CVPR)*, *2022*

Hanqing Wang, Wenguan Wang, Wei Liang and Jianbing Shen, Structured Scene Memory for Vision-Language Navigation,

Conference on Computer Vision and Pattern Recognition (CVPR), *2021*

Hanqing Wang, Wei Liang and Lap-Fai Yu, Scene Mover: Automatic Move Planning for Scene Arrangement by Deep Reinforcement Learning, *ACM Transactions on Graphics (ToG)*, *2020*

Hanqing Wang, Wenguan Wang, Tianmin Shu, Wei Liang, and Jianbing Shen, Active Visual Information Gathering for Vision-Language Navigation, *Proceedings of the European conference on Computer Vision (ECCV)*, *2020*

Hanqing Wang, Jiaolong Yang, Wei Liang and Xin Tong, Deep Single-View 3D Object Reconstruction with Visual Hull Embedding, *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, *2019*

Hanqing Wang, Wei Liang and Lap-Fai Yu, Transferring Objects: Joint Inference of Container and Human Pose, *International Conference on Computer Vision (ICCV)*, *2017*

GRANTS & LEADING PROJECTS

Shanghai Oriental Elite Program (Leading Project, PI) Shanghai Municipal Human Resources and Social Security Bureau

AWARDS

Shanghai Post-doctoral Excellence Program *Dec 2024* Shanghai Municipal Human Resources and Social Security Bureau

Beijing Outstanding Doctoral Graduate (Top 5%) *Jun 2024* Beijing Municipal Education Commission

Outstanding Doctoral Dissertation *Jun 2024* Beijing Institute of Technology

3rd Prize of Innovation Scholarship *Mar 2017* Ministry of Industry and Information Technology of the People's Republic of China

Award of Most Creative *Sep 2017* Microsoft Hackathon 2017, Microsoft

Award of Excellence *Nov 2017* Stars of Tomorrow Internship Program, Microsoft Research Asia

3rd Place in the HABITAT Challenge 2020 *Jun 2020* Facebook AI Research

National Scholarship *Oct 2021* Ministry of Education of the People's Republic of China

PROFESSIONAL SERVICES

Workshop & Challenge Organization

- Core Organizer, *IROS 2025 Workshop on Multimodal Robot Learning in Physical Worlds* [link]
- Core Organizer, *Challenge on Multimodal Robot Learning in InternUtopia and Real World*, 2025 [link]

Reviewing

- *Conferences:* CVPR, ECCV, ICCV, NeurIPS, AAAI, IJCAI, TVCG, SIGGRAPH
- *Journals:* IEEE T-PAMI, IJCV, IEEE TNNLS, IEEE TCSVT, IEEE VR, PR